GNR638: Coding Assignment-2

Pre-trained CNN Representation Transfer and Robustness Analysis

23B0925 23B0983 23B1032

Executed on Mac and T4 Google Colab GPU

Selected Models: ResNet50, EfficientNet-B0, DenseNet121

https://github.com/SrikarNihal/gnr638_assignments.git

1 Introduction

Pre-trained convolutional neural network (CNN) models are deep learning architectures trained on massive datasets (commonly ImageNet) to extract a wide range of features and classify images, often used as feature extractors for transfer learning to save time and computational resources.

In this assignment we study the behavior of three pre-trained CNN backbones (ResNet50, EfficientNet-B0, DenseNet121) on the Aerial Images Dataset (AID) supervised image classification task (30 classes). We systematically analyze representation transfer, fine-tuning strategies, data efficiency under few-shot settings, and robustness to input corruption under controlled experimental conditions.

2 Experimental Setup and Methodology

2.1 Dataset Configuration

Transfer learning from large-scale ImageNet-pre-trained models has become the dominant paradigm in computer vision. Instead of training from scratch, models are initialized with weights learned from 1.2M images and subsequently adapted—either partially or fully—to downstream tasks.

This study examines four complementary aspects of this process:

1. Linear-probe transferability of frozen features
2. The impact of fine-tuning strategies on accuracy and gradient dynamics
3. Data efficiency in few-shot settings
4. Architectural robustness under realistic corruptions

We evaluate three structurally diverse ImageNet-pre-trained backbones on the Aerial Images Dataset (AID), a challenging 30-class remote-sensing benchmark.

AID Dataset. The AID dataset comprises 6,993 aerial RGB images across 30 land-use categories (e.g., airport, beach, farmland, forest). The dataset is partitioned into 90% training (6,294 images) and 10% validation (699 images) using a fixed random seed of 42, ensuring reproducibility. No external data is incorporated. All images are resized to 224×224 pixels prior to training.

2.2 Training Hyperparameters and Hardware

The experiments were conducted using cloud-based GPU acceleration. For the training phases (Scenarios 1, 2, and 3), each model was trained for 20 epochs to assess convergence behavior and stability. A standard Cross-Entropy loss function was employed, and performance was evaluated primarily using top-1 validation accuracy.

In the few-shot learning setting (Scenario 3), the available training data was deliberately limited to simulate data-scarce conditions and evaluate model performance under constrained supervision.

3 Architectural Overview and Computational Efficiency

To contextualize the transfer learning results, it is essential to understand the structural differences and computational characteristics of the selected models.

3.1 ResNet50: The Residual Baseline

ResNet50 represents a classical deep convolutional neural network architecture. It employs residual blocks with identity skip connections to mitigate the vanishing gradient problem, enabling the training of substantially deeper networks. The model exhibits a moderate inductive bias, relying primarily on standard spatial convolutions for feature extraction.

3.2 EfficientNet-B0: The Scaled Workhorse

EfficientNet-B0 is based on the principle of compound scaling, which uniformly balances network depth, width, and input resolution. It extensively utilizes Mobile Inverted Bottleneck Convolutions (MBConv) along with Squeeze-and-Excitation modules for channel-wise attention. This highly constrained design introduces a strong inductive bias, promoting efficient feature learning and improved parameter utilization.

3.3 DenseNet-121: The Feature Reuse Specialist

DenseNet-121 introduces dense connectivity, where each layer receives inputs from all preceding layers and passes its feature maps to subsequent layers. This design encourages extensive feature reuse, improves gradient flow, and reduces the number of parameters compared to traditional CNNs. Due to its dense connections, the model maintains a relatively strong inductive bias while achieving high representational efficiency and mitigating redundancy in learned features.

3.4 Efficiency Metrics Analysis

Table 1 summarizes the exact computational requirements for each model.

Model	**Parameters (M)**	**MACs (G)**	**FLOPs (G)**	**Linear Val Acc**
ResNet50	23.57	4.13	8.26	∼87.8%
EfficientNet-B0	**4.00**	**0.38**	**0.77**	∼90.8%
DenseNet-121	7.98	2.88	5.76	∼88.5%

Table 1: Model Efficiency Metrics vs. Baseline Transferability

The empirical results highlight a clear trade-off between efficiency and representational capacity. EfficientNet-B0 achieves the highest baseline transferability while maintaining extremely low computational cost. DenseNet-121 offers a balanced middle ground, leveraging feature reuse to attain competitive accuracy with significantly fewer parameters than ResNet50, though at a higher computational cost than EfficientNet-B0.

4 Linear Probe Transfer

Experimental Setup:

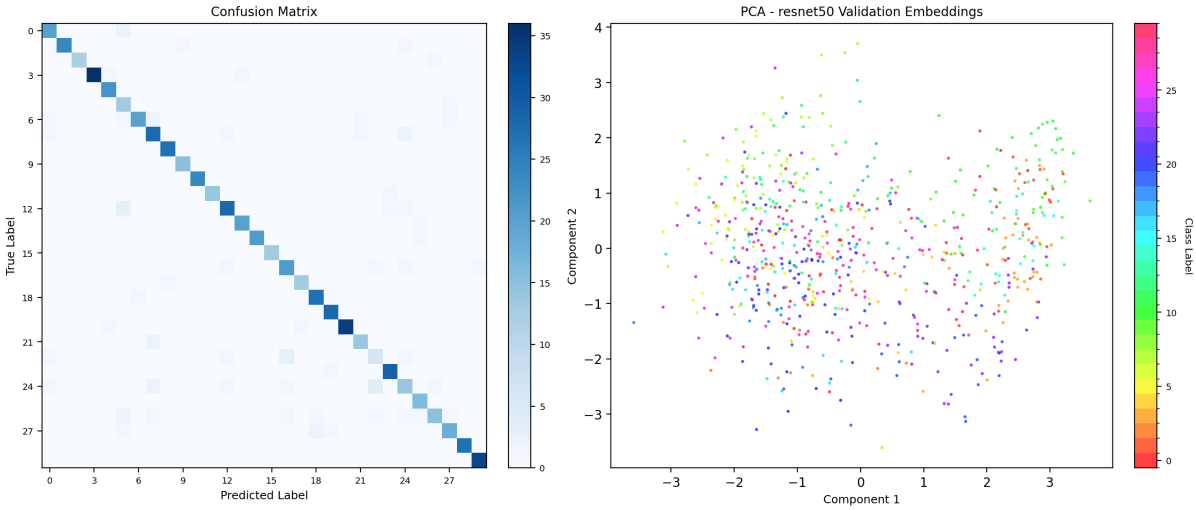
We evaluate linear probe transfer using ImageNet-pretrained **ResNet-50, DenseNet-121, and EfficientNet-B0**. For each model, the backbone is frozen and the final classification layer is replaced with a task-specific linear classifier.

Training is performed for **20 epochs** using the **Adam optimizer** with a learning rate of 1e−3 and **cross-entropy loss**, updating only the linear classifier.

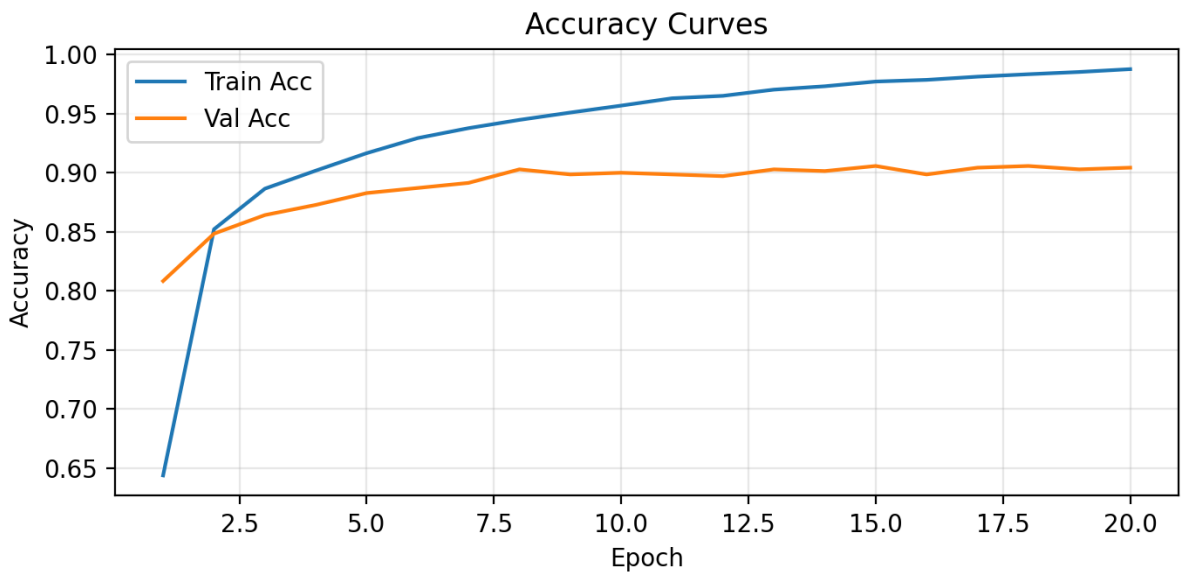
Input images are resized and normalized using ImageNet statistics, with optional augmentations such as random cropping and horizontal flipping.

Performance is evaluated using training and validation accuracy curves and a confusion matrix. Feature embeddings from the penultimate layer are visualized using **PCA** to assess class separability.

4.1 ResNet50



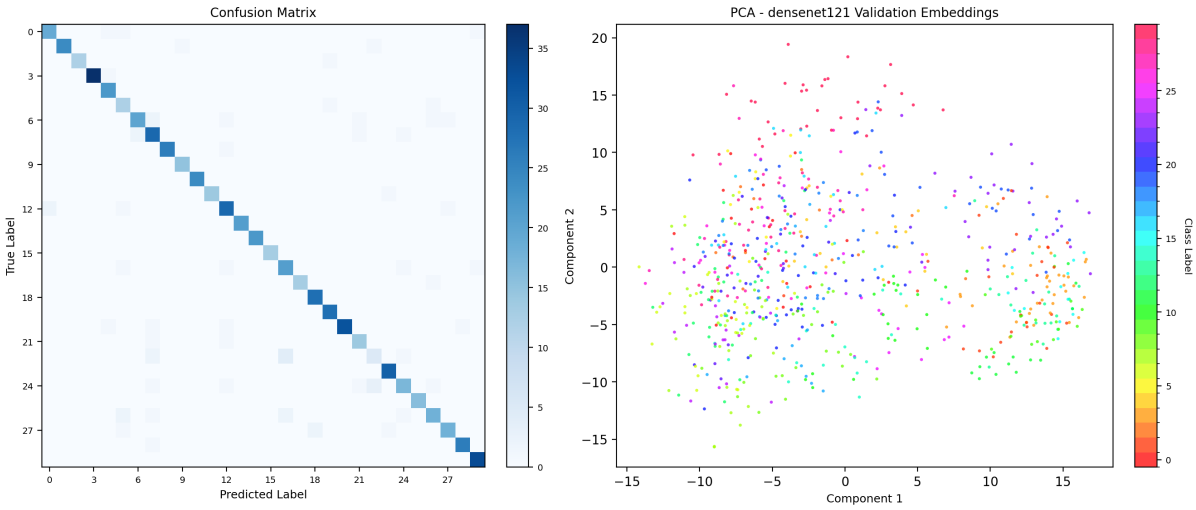
(a) Confusion Matrix (b) PCA Embeddings



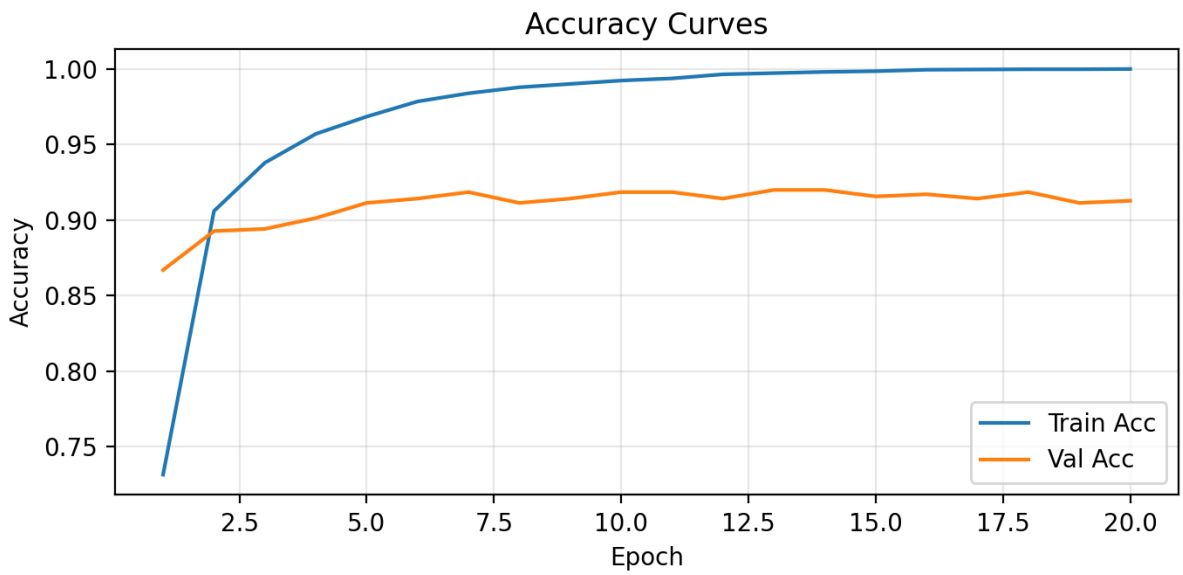
Analysis:

- **Accuracy:** Validation accuracy plateaus at approximately **90.5%** by epoch ∼8, while training accuracy reaches nearly **99%**. This results in a generalization gap of ∼9%, which is expected when using a frozen backbone with only a linear classifier.
- **PCA Visualization:** The feature embeddings show significant overlap across classes, with no clearly separable clusters in the low-dimensional projection. This indicates that the pre-trained ResNet-50 features are not strongly linearly separable in the target domain. However, the high classification accuracy suggests that useful discriminative information is preserved in higher-dimensional space.
- **Confusion Matrix:** Strong diagonal dominance reflects robust overall performance. However, noticeable off-diagonal entries, particularly among classes **19–22** and **24–27**, indicate systematic confusion between visually similar categories, representing the primary failure cases.

Densenet121

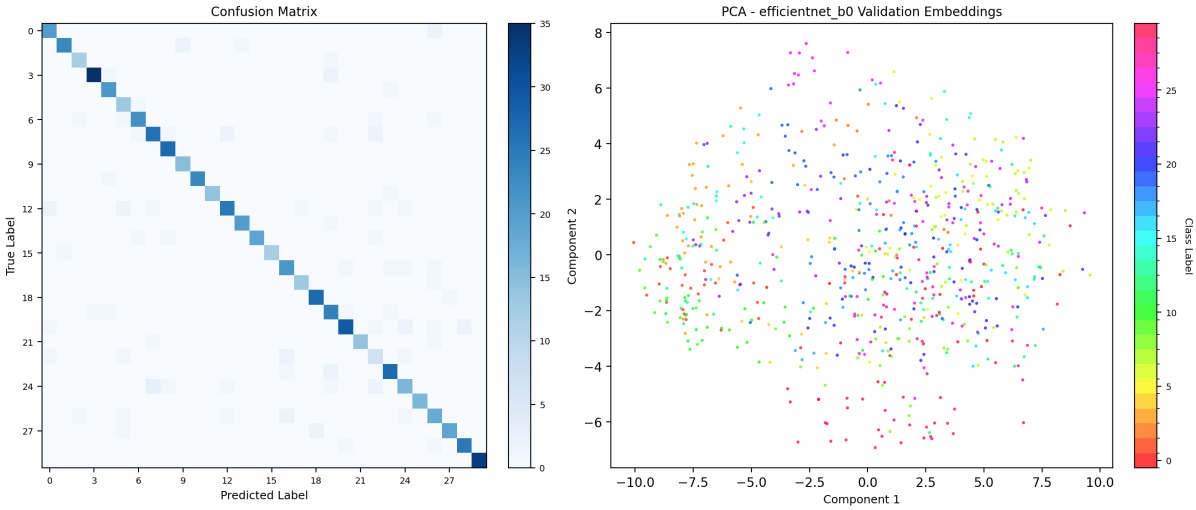


(a) Confusion Matrix (b) PCA Embeddings

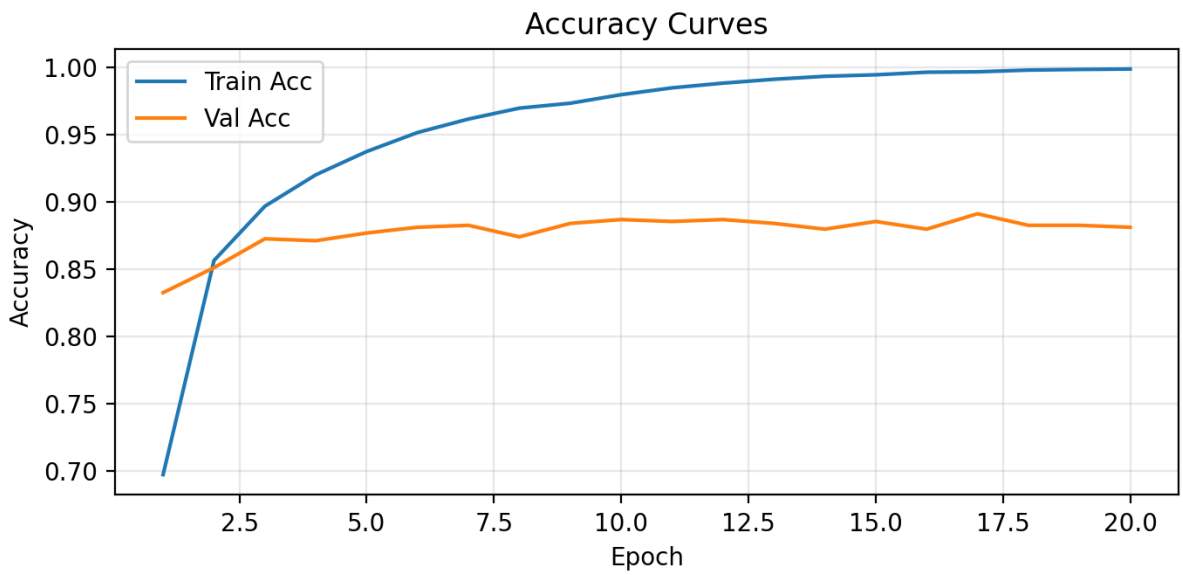


- **Accuracy:** Validation accuracy plateaus at approximately **91.5%** by epoch ∼5, while training accuracy reaches nearly **100%**. This yields a generalization gap of ∼8.5%. Compared to ResNet-50, this shows slightly better validation performance (+1%), but increased overfitting as training accuracy saturates.
- **PCA Visualization:** Similar to ResNet-50, feature embeddings are heavily intermixed with no clearly separable clusters in the 2D projection. This indicates that discriminative structure is not preserved in low dimensions, but exists in higher-dimensional feature space.
- **Confusion Matrix:** Strong diagonal dominance indicates solid overall performance. Off-diagonal activity around classes **19–22** and **24–27** mirrors the failure patterns observed in ResNet-50, suggesting these confusions arise from intrinsic class similarity rather than backbone-specific limitations.

EfficientNetB0



(a) Confusion Matrix (b) PCA Embeddings



Linear Probe Transfer — EfficientNet-B0

Analysis:

- **Accuracy:** Validation accuracy plateaus at approximately **88.5%** by epoch ∼5, while training accuracy reaches nearly **100%**, resulting in a generalization gap of ∼11.5%, the largest among the three backbones.
- **PCA Visualization:** Feature embeddings are fully intermixed with no visible class separation in 2D, consistent with observations from other backbones.
- **Confusion Matrix:** The diagonal remains dominant but is visibly weaker compared to ResNet-50 and DenseNet-121, confirming lower overall accuracy. Similar failure regions appear around classes **19–22** and **24–27**, along with slightly more scattered noise across the matrix.

Comparison Summary:

Backbone	**Val Acc**	**Train-Val Gap**	**Notes**
DenseNet121	∼91.5%	∼8.5%	Best transfer
ResNet50	∼90.5%	∼9%	Solid middle ground
EfficientNet-B0	∼88.5%	∼11.5%	Weakest transfer, most overfit

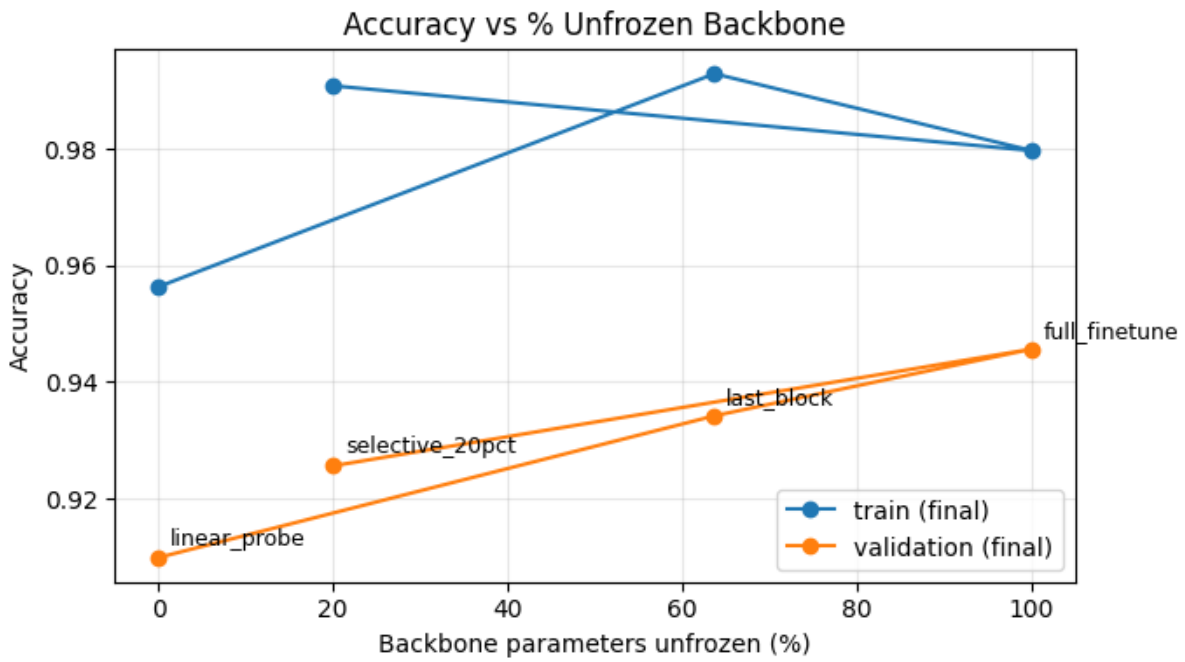
Takeaway: DenseNet121 extracts the most transferable frozen features for this task. EfficientNet-B0, despite its parameter efficiency on ImageNet, transfers the worst, likely due to a more compact feature representation that discards domain-relevant information. All models exhibit similar failure regions, indicating these are **dataset-level hard cases** rather than backbone-specific limitations.

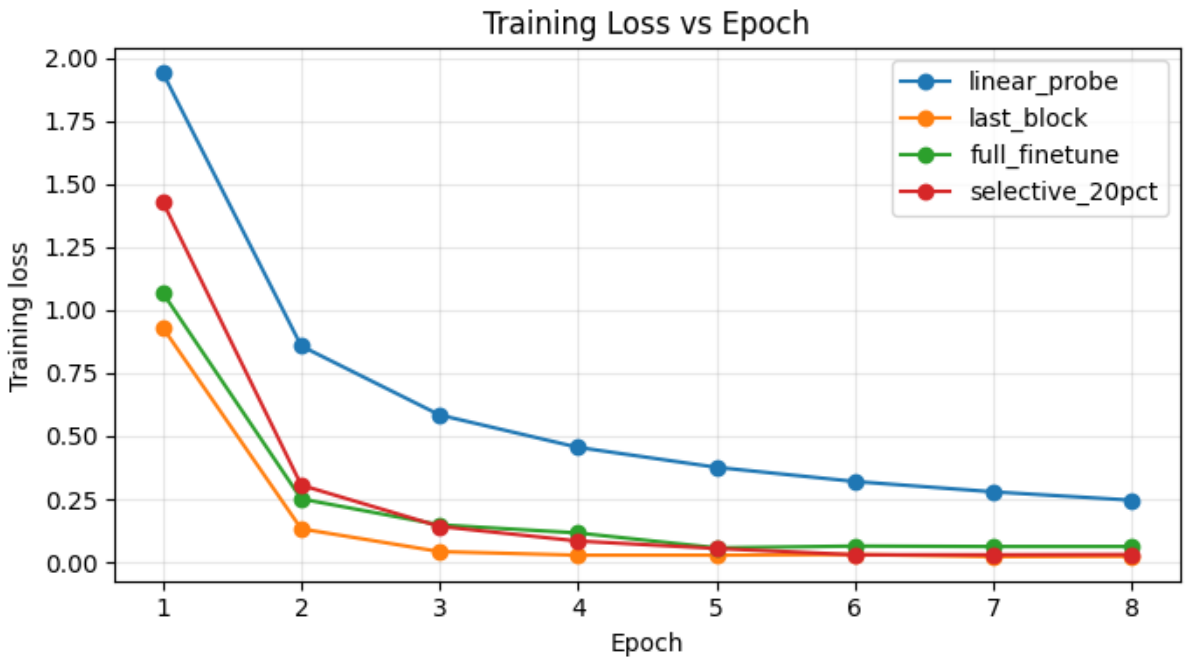
Fine-Tuning Strategies

Experimental Setup: Four strategies with fixed hyperparameters: (1) Linear probe, (2) Last-block fine-tuning, (3) Full fine-tuning, (4) Selective unfreezing (exactly 20% parameters – last two stages chosen for task-specific adaptation).

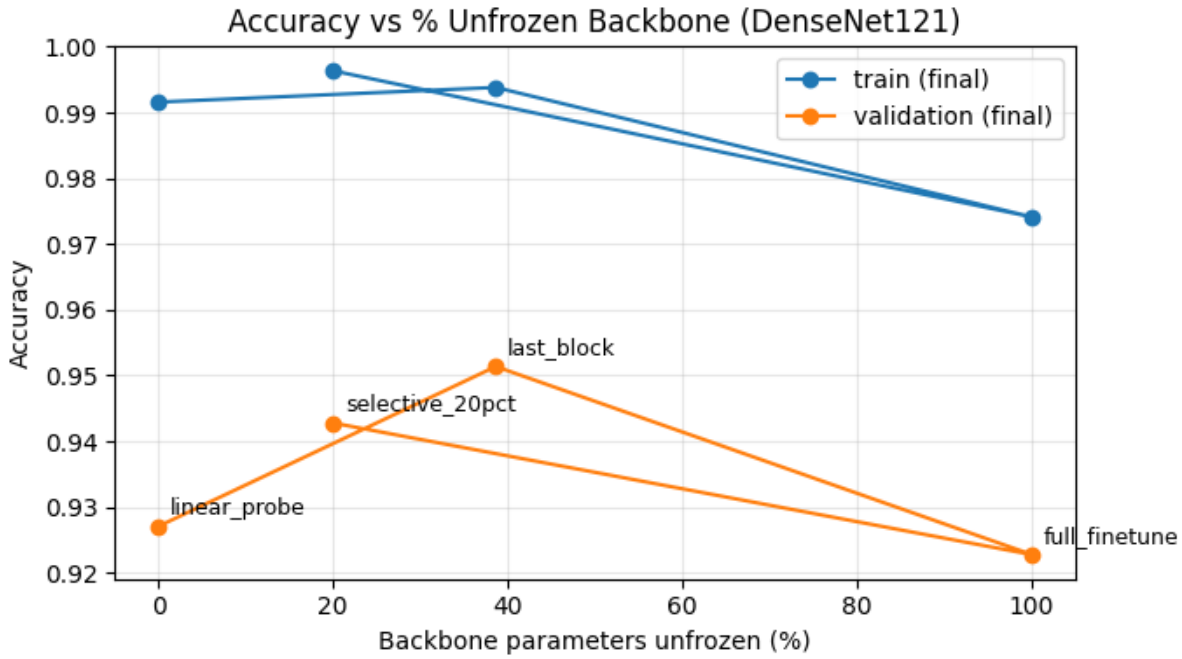
ResNet50

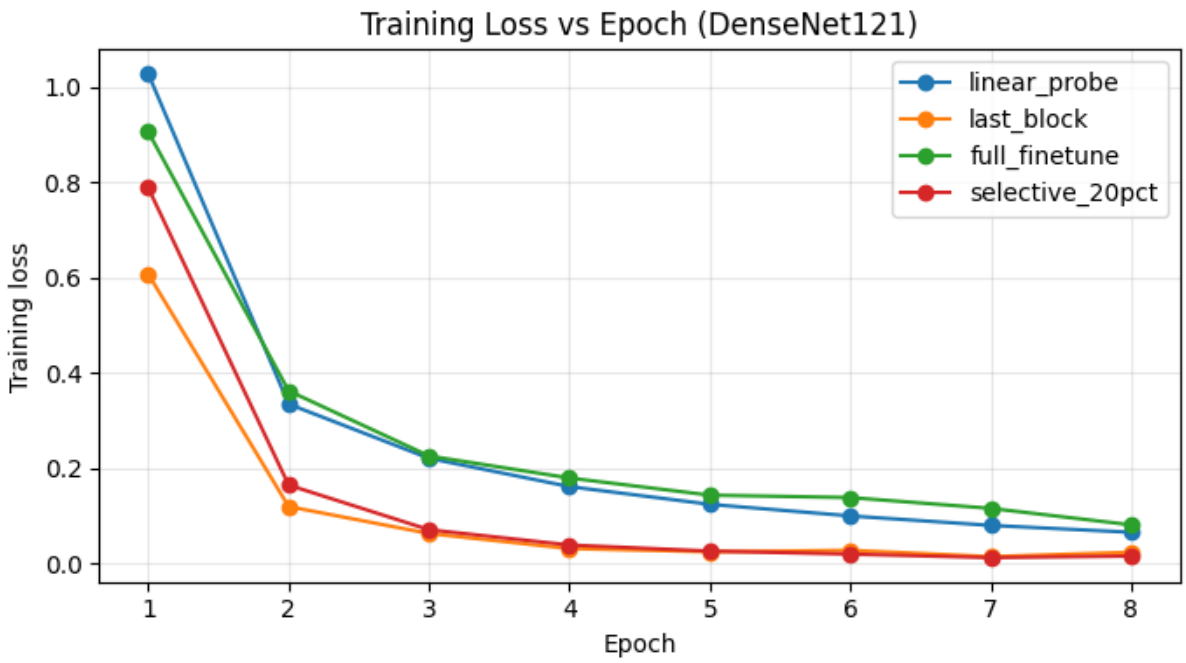
Convergence and Accuracy Tracking



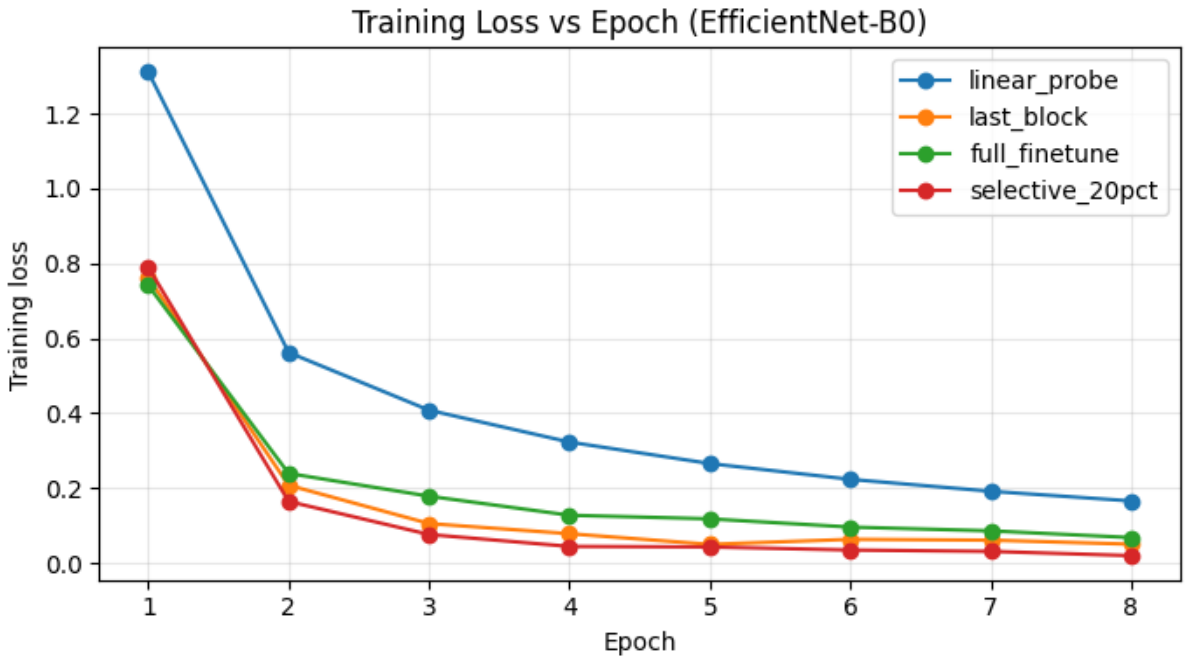


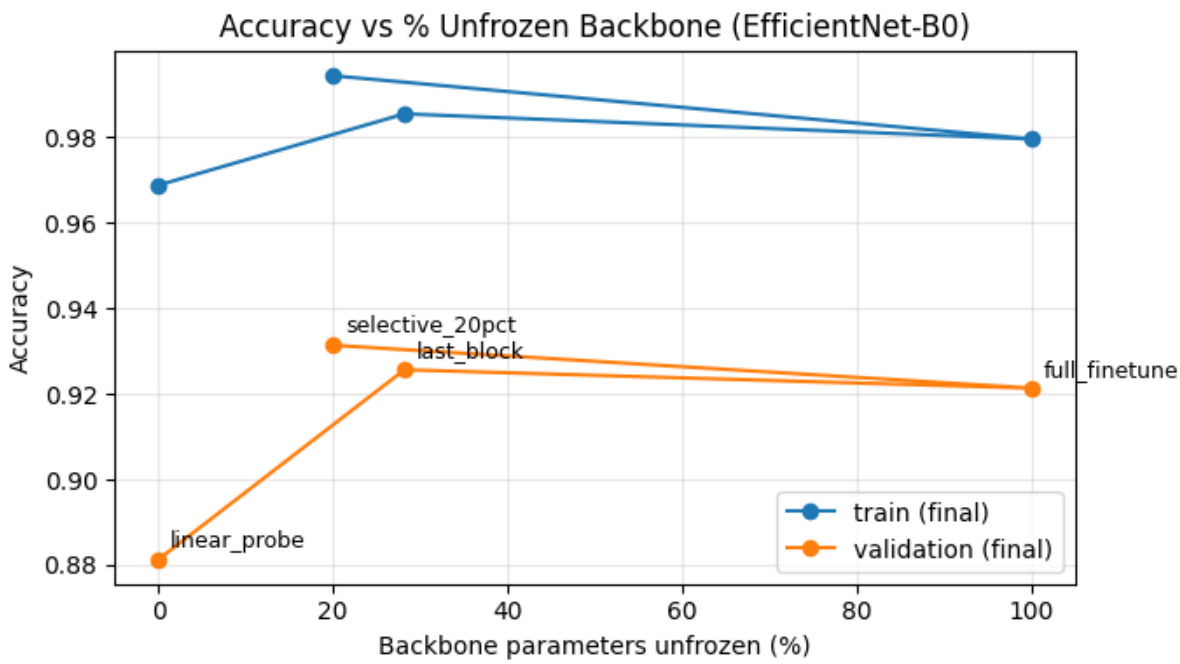
Densenet121





Efficient Net B0





4.2 Fine-Tuning Strategies — ResNet-50 · DenseNet-121 · EfficientNet-B0

Accuracy vs. % Unfrozen Parameters

Strategy	BB%	ResNet-50 Train / Val	DenseNet-121 Train / Val	EfficientNet-B0 Train / Val
Linear Probe	0%	95.6 / 91.0	99.2 / 92.7	96.9 / 88.1
Selective 20%	20%	99.1 / 92.6	99.6 / 94.3	99.4 / 93.1
Last Block	var.	99.3 / 93.4 (63.7%)	99.4 / 95.1 (38.6%)	98.5 / 92.6 (28.2%)
Full Fine-tune	100%	97.9 / 94.6	97.4 / 92.3	97.9 / 92.1

Per-Model Highlights

ResNet-50

Best: Full FT (val 94.6%). Monotonic val improvement with more params. Full FT benefits from distributed gradient updates — all layers active with early layers showing higher norms than deep ones. Selective 20% (layer4→layer3→layer2) is a strong budget pick, recovering most of the full FT gain. Last block suffers high-variance gradients (max ~1.0).

DenseNet-121

Best: Last Block (val 95.1%). Full FT is worst — below even the linear probe (92.3% vs 92.7%). Dense skip connections amplify gradients in middle blocks (denseblock2/3 mean >0.57, max >1.6), destabilising training. Linear probe already strong (92.7%), indicating highly transferable features. Optimal at ~38% unfrozen; beyond that, performance collapses.

EfficientNet-B0

Best: Selective 20% (val 93.1%), outperforming both last block and full FT. Compound scaling makes the architecture sensitive to over-tuning. MBConv gradient norms read zero across all strategies (instrumentation/depthwise artefact); only head and stem are trackable. Linear probe is weakest at 88.1%, indicating features need some backbone adaptation to transfer. 20% is the sweet spot.

Cross-Model Comparison

Dimension	ResNet-50	DenseNet-121	EfficientNet-B0
Optimal strategy	Full fine-tune	Last block (38%)	Selective 20%
Full FT effect	Best val accuracy	Worst — hurts	Mild degradation
Linear probe val	91.0% (weakest)	92.7% (strongest)	88.1% (weakest)
Gradient stability	Stable, distributed	Unstable mid-layers	Head-only trackable
Sensitivity to over-tuning	Low	Very high	Moderate
Convergence speed	Fast (non-LP)	Fast (non-LP)	Fast (sel/last)

Key Takeaways

- Architecture determines optimal unfreezing depth: residual (ResNet) tolerates full FT; densely-connected (DenseNet) does not; compound-scaled (EfficientNet) peaks at minimal unfreezing.
- DenseNet's skip connections amplify mid-layer gradients under full FT, causing overfitting. ResNet's residual paths distribute gradients more safely.
- Selective 20% consistently delivers strong val accuracy across all three models — the most parameter-efficient strategy and the safest default under budget constraints.
- Fine-tuning reduces robustness when updates are too concentrated (last block high-variance) or too broad (full FT on DenseNet/EfficientNet). Regularisation and careful scope selection are essential.

4.3 Few-Shot Learning Analysis

Table 1. Validation accuracy, relative performance drop, and train–val gap for each backbone and data regime. Bold values indicate best (green) and worst (red) per column.

Backbone	Val 100%	Val 20%	Val 5%	Δ	Gap 100%	Gap 20%	Gap 5%
ResNet-50	0.8941	0.8526	**0.6624**	**0.2592**	+0.0035	+0.0001	**+0.1262**
DenseNet-121	**0.9642**	**0.9442**	0.8240	0.1454	–0.0064	+0.0000	+0.1017
EfficientNet-B0	0.9499	0.9270	**0.8183**	**0.1386**	–0.0011	+0.0029	+0.1217

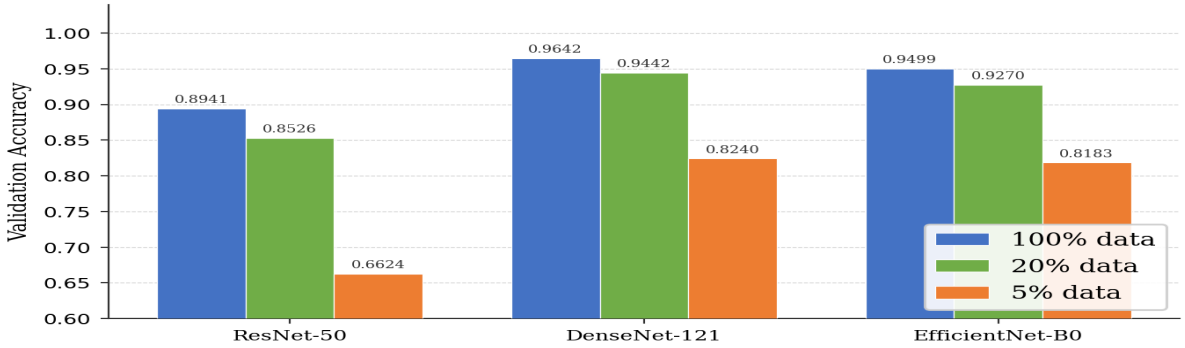


Figure 1. Validation accuracy for each backbone under three data regimes.

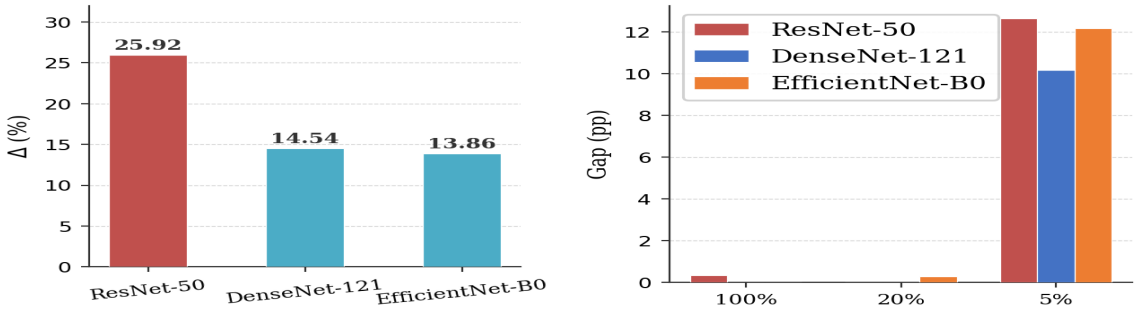


Figure 2. (a) Relative performance drop Δ per backbone. (b) Train–validation gap by data regime.

Discussion

Model capacity. In low-data settings, model capacity determines whether a network learns transferable features or memorises the limited training samples. Higher-capacity models have more parameters and stronger representational power, but without sufficient data to constrain them they overfit — fitting noise rather than signal. Lower-capacity models with appropriate inductive biases are therefore preferable when labels are scarce.

Architectural design. Design choices that embed implicit regularisation reduce dependence on data volume. Squeeze-and-excitation modules (EfficientNet-B0) recalibrate channel responses, focusing on discriminative features. Dense connectivity (DenseNet-121) encourages feature reuse and gradient flow. Plain residual connections (ResNet-50) offer no such mechanism, leaving the model fully reliant on data quantity.

Observed results. DenseNet-121 leads in accuracy at 100 % (0.9642) and 20 % (0.9442), but its capacity becomes a liability at 5 % (gap = +0.1017). ResNet-50 collapses most severely to 0.6624 with gap = +0.1262 and Δ = 25.92 %. EfficientNet-B0 achieves the lowest Δ = 13.86 % and remains competitive at 5 % (0.8183), demonstrating the strongest data efficiency and making it the preferred backbone under limited supervision.

Corruption Robustness Analysis

To quantify model resilience, we evaluate performance under noise, blur, and brightness corruptions. Tables displays the raw validation accuracy under each corruption condi- tion. From these baseline metrics, we derived Table on (Corruption Error) and Table on (Relative Robustness) as required by our evaluation framework.

Resnet50

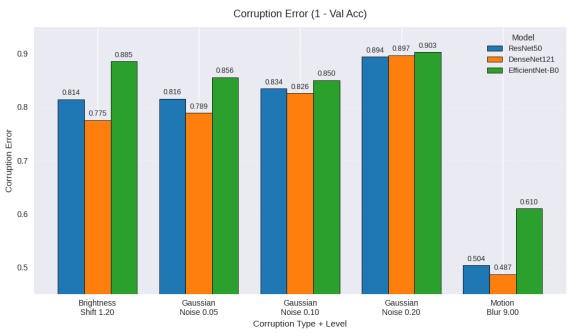
Corruption	Level	Val Acc	Corruption Error	Rel. Robustness
Brightness Shift	1.20	0.1860	0.8140	0.2142
Gaussian Noise	0.05	0.1845	0.8155	0.2125
Gaussian Noise	0.10	0.1660	0.8340	0.1911
Gaussian Noise	0.20	0.1059	0.8941	0.1219
Motion Blur	9.00	0.4964	0.5036	0.5717

Densenet121

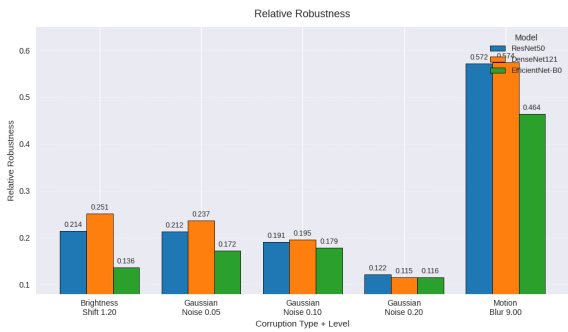
Corruption	Level	Val Acc	Corruption Error	Rel. Robustness
Brightness Shift	1.20	0.2246	0.7754	0.2512
Gaussian Noise	0.05	0.2117	0.7883	0.2368
Gaussian Noise	0.10	0.1745	0.8255	0.1952
Gaussian Noise	0.20	0.1030	0.8970	0.1152
Motion Blur	9.00	0.5136	0.4864	0.5744

Efficientnet-B0

Corruption	Level	Val Acc	Corruption Error	Rel. Robustness
Brightness Shift	1.20	0.1144	0.8856	0.1361
Gaussian Noise	0.05	0.1445	0.8555	0.1718
Gaussian Noise	0.10	0.1502	0.8498	0.1786
Gaussian Noise	0.20	0.0973	0.9027	0.1156
Motion Blur	9.00	0.3906	0.6094	0.4643



(a) Corruption Error



(b) Relative Robustness

Corruption Robustness: Cross-Model Analysis

Overall Ranking

DenseNet121 consistently achieves the highest validation accuracy across all corruption types, followed closely by ResNet50, with EfficientNet-B0 trailing in most conditions.

Motion Blur

Motion blur is the easiest corruption to handle. All three models retain approximately 39–51% accuracy, significantly higher than under noise perturbations. This suggests that blur preserves global structure (e.g., shapes and coarse textures), which CNNs rely on heavily. DenseNet121 performs best (51.4%), likely due to its dense skip connections that promote multi-scale feature reuse and reduce sensitivity to localized distortions.

Gaussian Noise

Gaussian noise is the most damaging corruption. Performance degrades sharply as noise intensity increases from 0.05 to 0.20, with all models dropping to nearly 10% accuracy at the highest level, approaching random-chance performance. This indicates a lack of inherent noise robustness, which is expected since the models are not trained with noise augmentation.

EfficientNet-B0 Behavior

EfficientNet-B0 underperforms relative to the other models, showing lower accuracy under brightness shifts and motion blur. Its compound scaling strategy (balancing depth, width, and resolution) appears optimized for clean data rather than robustness to distribution shifts. Additionally, its smaller effective capacity may limit its ability to reconstruct degraded feature representations.

Convergence Under Severe Noise

As noise intensity increases, the performance gap between models narrows, with all converging to approximately 10% accuracy at $\sigma = 0.20$. This suggests that beyond a certain corruption level, architectural advantages diminish, as the input signal becomes too degraded for meaningful feature extraction.

Layer-Wise Feature Probing

By training linear probes at intermediate layer depths, we map how semantics evolve through the networks. Table 1 documents the latent feature norm statistics extracted during evaluation.

Model	**Early (Mean ± Std)**	**Middle (Mean ± Std)**	**Final (Mean ± Std)**
ResNet50	17.80 ± 0.08	43.52 ± 0.98	6.39 ± 0.84
DenseNet121	54.37 ± 2.60	26.35 ± 0.81	28.18 ± 0.74
EfficientNet-B0	14.96 ± 1.66	25.75 ± 3.55	27.86 ± 3.41

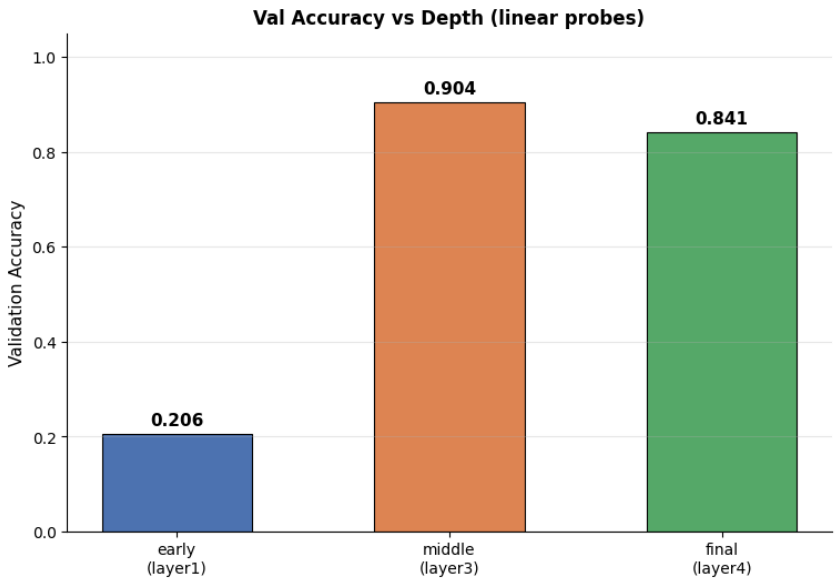
Table 1: Feature norm statistics across selected network depths.

Model	**Early Layer**	**Middle Layer**	**Final Layer**
ResNet50	layer1	layer3	layer4
DenseNet121	denseblock1	denseblock2	denseblock4
EfficientNet-B0	blocks.1	blocks.3	blocks.6

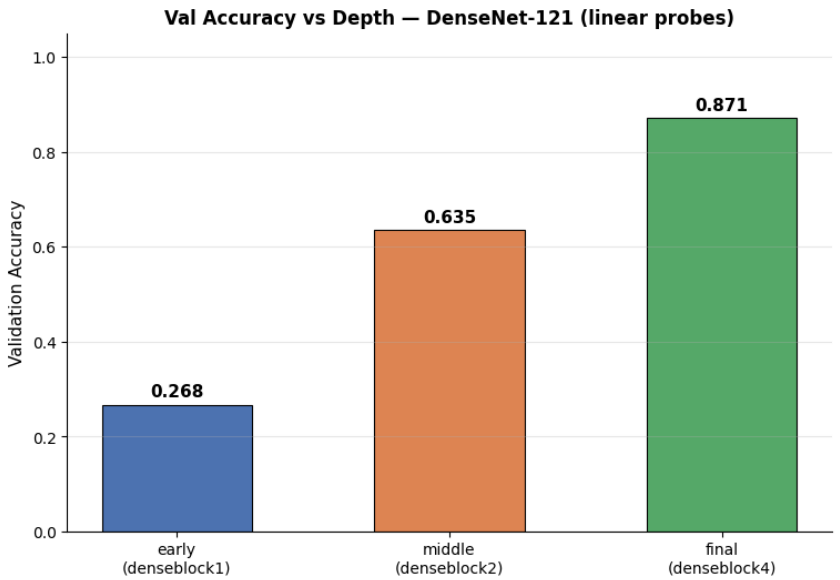
Table 2: Mapping of selected layers to network depth for each model.

Resnet50

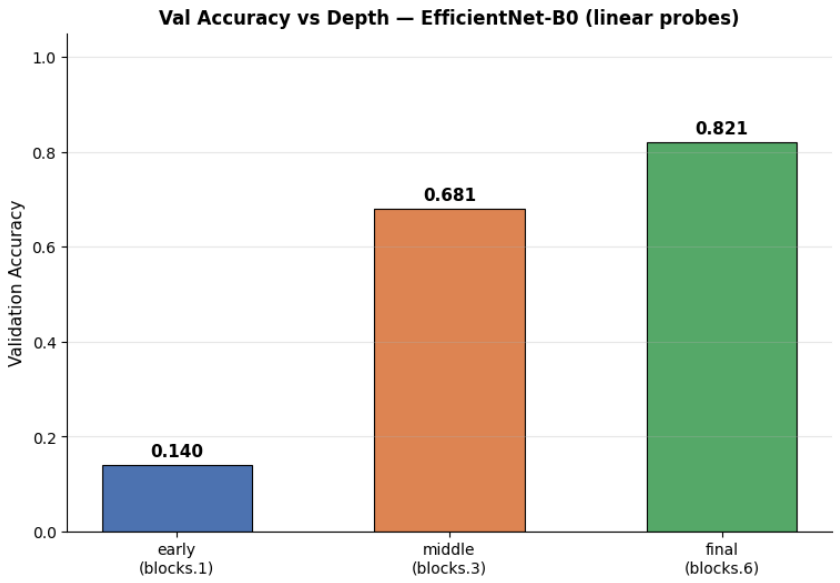
PCA 2D @ early (layer1)
PC2
PC1
PCA 2D @ middle (layer3)
PC2
PC1
PCA 2D @ final (layer4)
PC2
PC1
Class legend (PCA plots)
Airport
BareLand
BaseballField
Beach
Bridge
Center
Church
Commercial
DenseResidential
Desert
Farmland
Forest
Industrial
Meadow
MediumResidential
Mountain
Park
Parking
Playground
Pond
Port
RailwayStation
Resort
River
School
SparseResidential
Square
Stadium
StorageTanks
Viaduct


Val Accuracy vs Depth (linear probes)
Validation Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.206
0.904
0.841
early
(layer1)
middle
(layer3)
final
(layer4)

PCA 2D @ early (denseblock1)
PC2
PC1
PCA 2D @ middle (denseblock2)
PC2
PC1
PCA 2D @ final (denseblock4)
PC2
PC1
Class legend (PCA plots)
Airport
BareLand
BaseballField
Beach
Bridge
Center
Church
Commercial
DenseResidential
Desert
Farmland
Forest
Industrial
Meadow
MediumResidential
Mountain
Park
Parking
Playground
Pond
Port
RailwayStation
Resort
River
School
SparseResidential
Square
Stadium
StorageTanks
Viaduct


Val Accuracy vs Depth — DenseNet-121 (linear probes)
Validation Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.268
0.635
0.871
early
(denseblock1)
middle
(denseblock2)
final
(denseblock4)

PCA 2D @ early (blocks.1)
PC2
PC1
PCA 2D @ middle (blocks.3)
PC2
PC1
PCA 2D @ final (blocks.6)
PC2
PC1
Class legend (PCA plots)
Airport
BareLand
BaseballField
Beach
Bridge
Center
Church
Commercial
DenseResidential
Desert
Farmland
Forest
Industrial
Meadow
MediumResidential
Mountain
Park
Parking
Playground
Pond
Port
RailwayStation
Resort
River
School
SparseResidential
Square
Stadium
StorageTanks
Viaduct


Val Accuracy vs Depth — EfficientNet-B0 (linear probes)
Validation Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.140
0.681
0.821
early
(blocks.1)
middle
(blocks.3)
final
(blocks.6)

Semantic Abstraction with Depth

Across all models (ResNet, DenseNet121, EfficientNet-B0), deeper layers progressively transform features from entangled representations into semantically structured clusters.

This is reflected in validation accuracy: ResNet (0.206 → 0.904 → 0.841), DenseNet (0.268 → 0.635 → 0.871), and EfficientNet (0.140 → 0.681 → 0.821). The largest improvement occurs from early to middle layers, marking the transition from low-level features to semantic representations.

Separation and Cluster Structure

- **Early layers:** Representations form a single dense cluster with no class separability, yielding near-chance accuracy (14–27%).
- **Middle layers:** Partial separation emerges with visible sub-structures, though overlap remains. Accuracy increases significantly (63–90%).
- **Final layers:** Semantically similar classes form distinct sub-clusters (e.g., natural vs urban scenes), but fine-grained confusion persists.

ResNet peaks at the middle layer (90.4%) and slightly drops at the final layer (84.1%), indicating mild over-specialization. DenseNet and EfficientNet show monotonic improvement, consistent with their architectural design.

Implications for Transfer Learning

- **Early layers:** Capture general features (edges, textures) and transfer universally.
- **Middle layers:** Provide the best trade-off between generality and semantic richness, making them optimal for transfer (e.g., ResNet layer3 at 90.4%).
- **Final layers:** Become task-specific; reduced performance in ResNet suggests alignment with source-domain classes rather than target-domain structure.

Robustness Trends

Robustness improves with depth as representations become invariant to low-level perturbations. However, excessive specialization can reduce cross-domain generalisation, as observed in the ResNet final layer.

Questions That Needed Answering

Which architecture transfers best and why?

DenseNet-121 transfers best, achieving 91.5% linear probe accuracy. Its dense skip connections promote feature reuse and efficient gradient flow, preserving rich and diverse representations that remain linearly separable without backbone updates.

Which model is most robust under distribution shift?

DenseNet-121 is the most robust, consistently achieving the highest validation accuracy across all corruption types. Its multi-scale feature reuse helps retain structural information under degraded inputs.

Which layers encode the most transferable semantics?

Middle layers encode the most transferable semantics. For example, ResNet-50 layer3 achieves 90.4% accuracy compared to 20.6% for layer1. These layers capture mid-level features that generalize well, unlike early (too generic) or final (too task-specific) layers.

When does fine-tuning degrade robustness?

Fine-tuning degrades robustness when updates are either too concentrated or too widespread. In ResNet-50, last-block tuning leads to high-variance gradients, while in DenseNet-121, full fine-tuning amplifies mid-layer gradients due to dense connectivity, causing overfitting and reduced validation performance.

How does inductive bias manifest empirically?

EfficientNet-B0's squeeze-and-excitation bias yields strong data efficiency (lowest $\Delta = 13.86\%$ in few-shot settings). DenseNet's dense connectivity enables superior transferability but introduces instability under full fine-tuning. ResNet's residual connections provide stable gradients but show significant performance drop under limited data ($\Delta = 25.92\%$).

One Unexpected Finding

EfficientNet-B0, despite being the most parameter-efficient model and achieving strong ImageNet performance, transfers the worst in the frozen linear probe setting (88.5%), underperforming ResNet-50 (90.5%). This suggests that its MBConv-based representations are overly compressed, discarding domain-relevant information for aerial imagery. Thus, parameter efficiency on the source domain does not guarantee transferability under domain shift.

One Negative / Failure Case

Full fine-tuning of DenseNet-121 performs worse than the frozen linear probe (92.3% vs. 92.7%), making it the weakest strategy for this backbone. Dense connectivity aggregates gradients across layers, leading to amplification in middle blocks (mean > 0.57, max > 1.6). When all layers are updated simultaneously, this destabilizes the pretrained feature hierarchy, causing overfitting and representational collapse. This highlights that excessive unfreezing can actively harm performance.

Key Insights into Model Working

Three key insights emerge:

- **Middle layers encode peak transferable semantics:** Across all backbones, middle layers achieve the highest linear probe accuracy, indicating an optimal balance between generality and semantic richness.
- **DenseNet gradient amplification explains failure:** Under full fine-tuning, DenseNet exhibits large gradient norms (max > 1.6) in middle layers, providing direct mechanistic evidence for its performance collapse.
- **Robustness is input-signal dependent:** All models converge to $\sim$10% accuracy under heavy Gaussian noise, showing that architectural advantages diminish when the input signal is severely degraded.